FEATURE BRIEF · TOWER X

QS Cost — what we built

Nineteen features that help a Quantity Surveyor see cost trouble one reporting period before it lands.

19 features | **39** API endpoints | **173** cost centres × 12 periods | **224.3M AED** under management

Full detail — worked examples, formulas, honest limits — in **QS-Cost-Feature-Guide.pdf**.

THE PROBLEM

A QS spots the overrun weeks too late.

- Two questions, forever: **what will this finally cost?** and **where are we drifting?**
- Both answered in a spreadsheet, at month-end, from numbers already weeks old.
- By the time a cost centre reads AMBER, **the money that caused it is spent**.

The line that matters

On Tower X a centre goes AMBER exactly when **CPI falls below 0.95** — the project's own rule, matching the supplied flag on every row.

The history holds **117** GREEN → AMBER transitions. Every one of them was visible the month before.

WHAT WE BUILT

A system of record, not a report.

19	**39**	**2,076**	**163**
features, wired end to end across 14 tabs	API endpoints over a multi-tenant Postgres store	panel rows — every EVM figure **derived**, never looked up	automated tests, including the arithmetic tie-outs

The workbook's computed EVM, KPI and progress sheets were deliberately withheld. Everything this system shows for Tower X had to be computed from the bill, the estimate and 12 months of history.

THE SHAPE OF IT

Seven jobs a QS actually does.

Where we stand 2

Live EVM dashboard · cost-centre grid and detail drawer

See trouble early 4

Watchlist · variance attribution · hindsight proof · model validation

Look forward 3

Spend forecaster with a real band · unit-rate what-if · correction actions

Before you award 1

Estimate assumption stress test — three separated output classes

Ask in English 1

QS Copilot — 13 read-only tools, every answer carries its evidence

On the building 4

3D Cost X-Ray · IFC take-off · element register · 4D build sequence

Run it as a system 4

Reporting workflow · projects · data administration · tenant isolation

Every feature says what it cannot prove

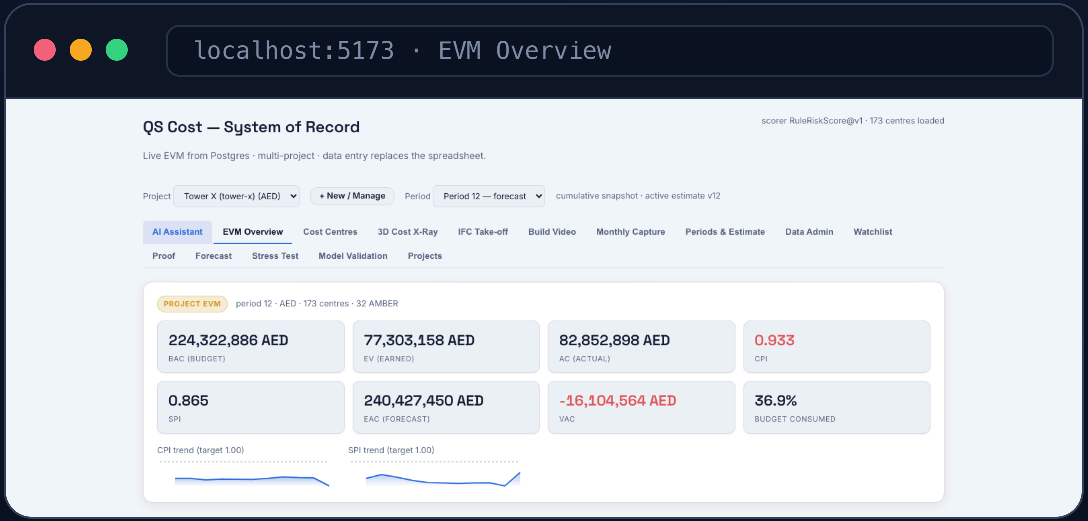
Directional figures are labelled on screen and listed in one honesty ledger at the back of the guide.

01 · WHERE THE PROJECT STANDS

The spreadsheet, computed.

- **Live EVM Dashboard** — *what is the project's cost health right now?* Totals and CPI/SPI trends for any period, computed on request.
- **Cost Centre Grid & Detail** — *which of 173 centres owns the problem?* Sortable, filterable, one click into the full picture.

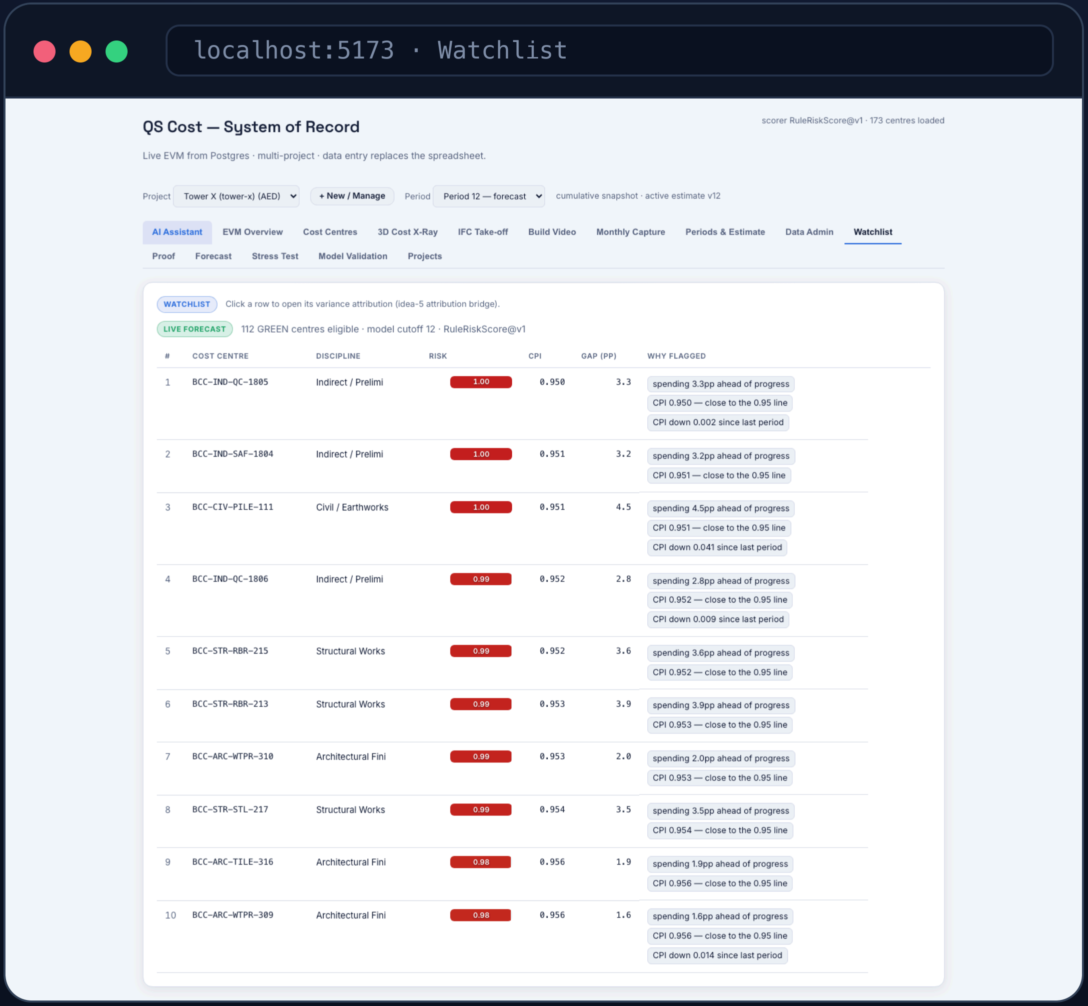
Project CPI is always Σ EV ÷ Σ AC — never the mean of per-centre ratios.



02 · SEE TROUBLE EARLY

A ranked triage list, **one period early.**

- **Early-Warning Watchlist** — *which GREEN centres are about to tip?* A published two-term score, not a black box.
- **Variance Attribution** — *where is the over-run coming from?* Split by resource, reconciled to the dirham.
- **Proof** — *would it have worked last month?* Rewind and let history grade it.
- **Model Validation** — *how accurate is it, honestly?*



#	COST CENTRE	DISCIPLINE	RISK	CPI	GAP (PP)	WHY FLAGGED
1	BCC-IND-QC-1805	Indirect / Prelimi	1.00	0.950	3.3	spending 3.3pp ahead of progress CPI 0.950 — close to the 0.95 line CPI down 0.002 since last period
2	BCC-IND-SAF-1804	Indirect / Prelimi	1.00	0.951	3.2	spending 3.2pp ahead of progress CPI 0.951 — close to the 0.95 line
3	BCC-CIV-PILE-111	Civil / Earthworks	1.00	0.951	4.5	spending 4.5pp ahead of progress CPI 0.951 — close to the 0.95 line CPI down 0.041 since last period
4	BCC-IND-QC-1806	Indirect / Prelimi	0.99	0.952	2.8	spending 2.8pp ahead of progress CPI 0.952 — close to the 0.95 line CPI down 0.009 since last period
5	BCC-STR-RBR-215	Structural Works	0.99	0.952	3.6	spending 3.6pp ahead of progress CPI 0.952 — close to the 0.95 line
6	BCC-STR-RBR-213	Structural Works	0.99	0.953	3.9	spending 3.9pp ahead of progress CPI 0.953 — close to the 0.95 line
7	BCC-ARC-WTPR-310	Architectural Fini	0.99	0.953	2.0	spending 2.0pp ahead of progress CPI 0.953 — close to the 0.95 line
8	BCC-STR-STL-217	Structural Works	0.99	0.954	3.5	spending 3.5pp ahead of progress CPI 0.954 — close to the 0.95 line
9	BCC-ARC-TILE-316	Architectural Fini	0.98	0.956	1.9	spending 1.9pp ahead of progress CPI 0.956 — close to the 0.95 line
10	BCC-ARC-WTPR-309	Architectural Fini	0.98	0.956	1.6	spending 1.6pp ahead of progress CPI 0.956 — close to the 0.95 line CPI down 0.014 since last period

THE NUMBER

We let the app grade itself.

45% vs **35%**

precision@5 — the deployed rule against the best CPI-native baseline

117

real GREEN → AMBER transitions, across 8 leakage-safe folds

2.8

false alerts per cycle — the cost of looking early

Published, not hidden

The per-fold range is **20%–80%** and it is on the screen next to the mean, because the mean alone would overstate how settled the number is.

The question we could not ask

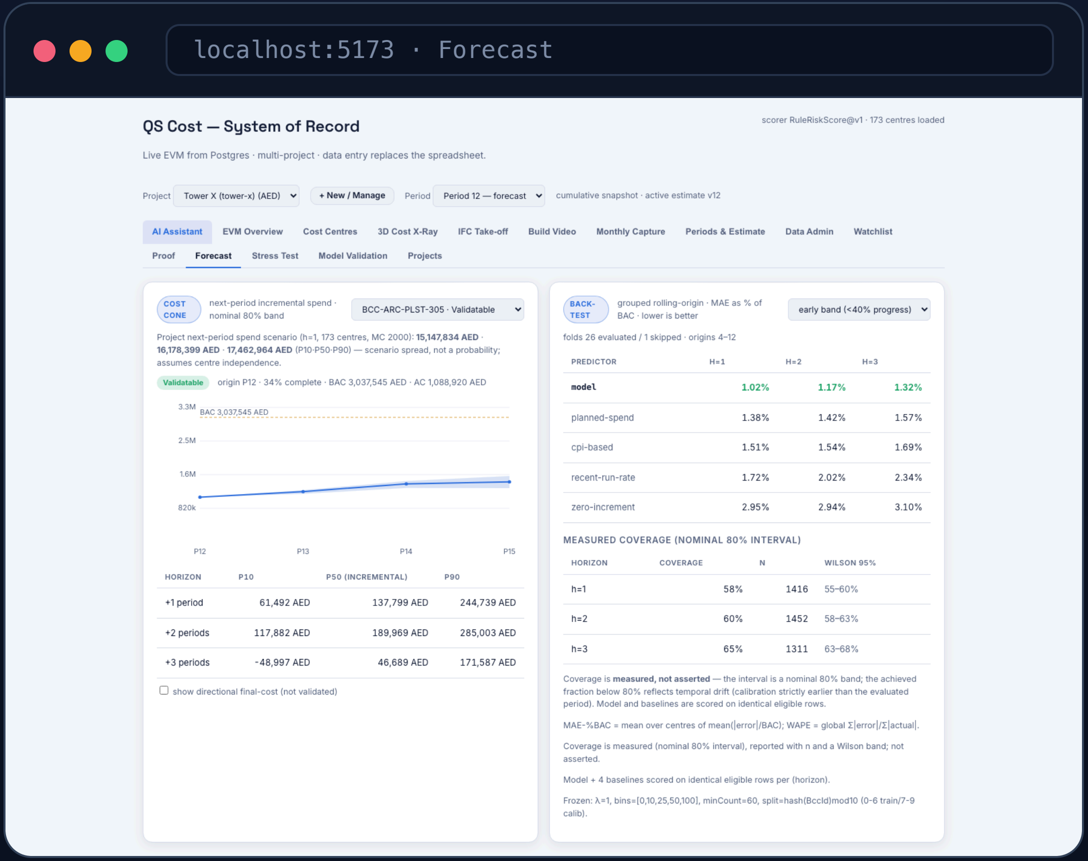
We set out to test whether physical neighbourhood predicts drift. On Tower X **no discipline spans more than one zone** — so a "spatial" feature would really measure trade. We published the finding instead of shipping the feature.

03 · LOOK FORWARD

Next-period spend, with a real band.

- **Cost-Trajectory Forecaster** — *how much next period, and what range?* P10–P90 from the model's own out-of-fold errors, not an assumed distribution.
- **Unit-Rate What-If** — *what is a renegotiation worth?*
- **Correction Actions** — *what could we do, and what would it save?*

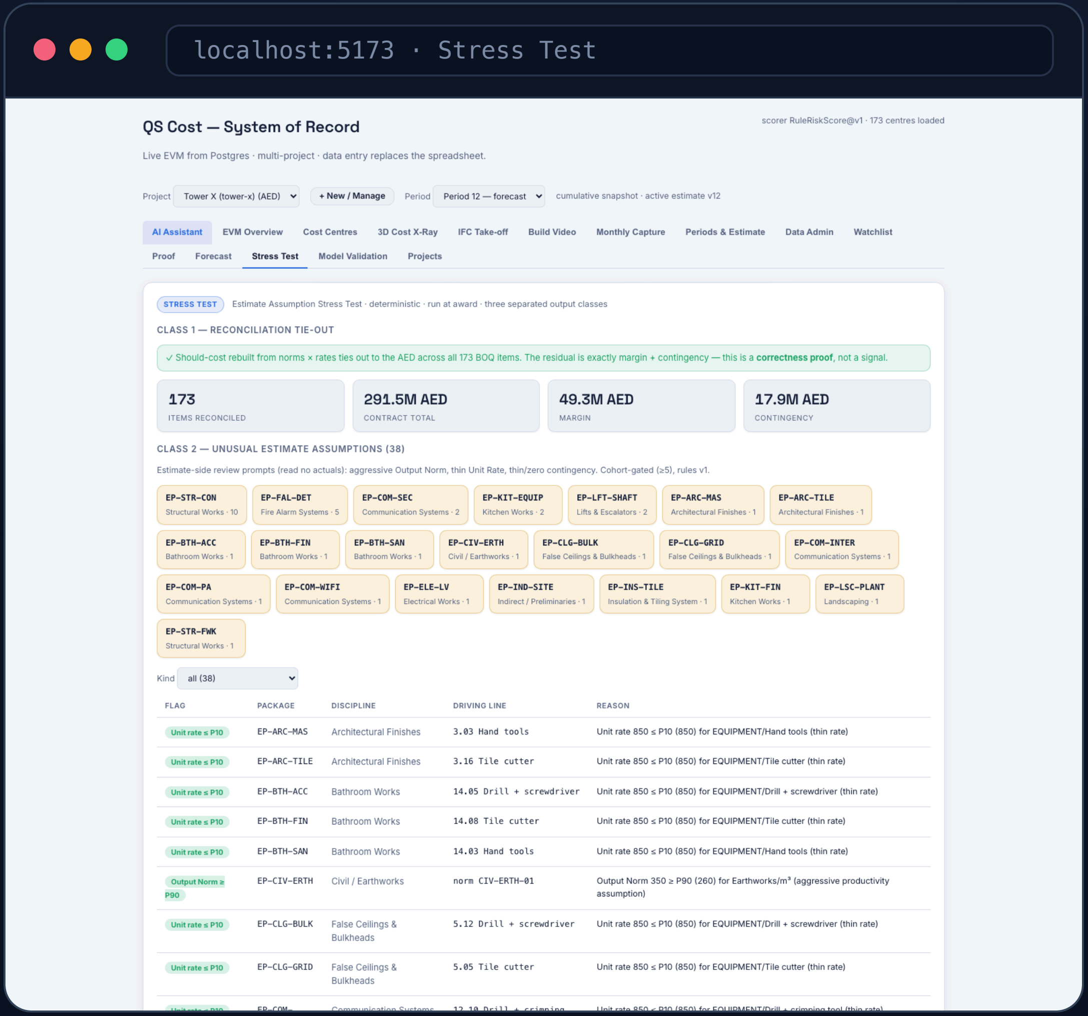
Coverage of the 80% band is **measured, not asserted**: 58% / 60% / 65% at h=1/2/3, published with sample sizes and confidence intervals.



04 · BEFORE YOU AWARD

Catch trouble before award.

- **Class 1 — reconciliation.** Should-cost rebuilt from norms × rates ties out to the AED across all 173 items. A correctness proof, not a signal.
- **Class 2 — assumption flags.** 38 cohort-gated review prompts: aggressive output norms, thin rates, zero contingency. Reads no actuals.
- **Class 3 — peer benchmark.** Needs 5 completed peers. Tower X has none, so it is **suppressed rather than faked**.

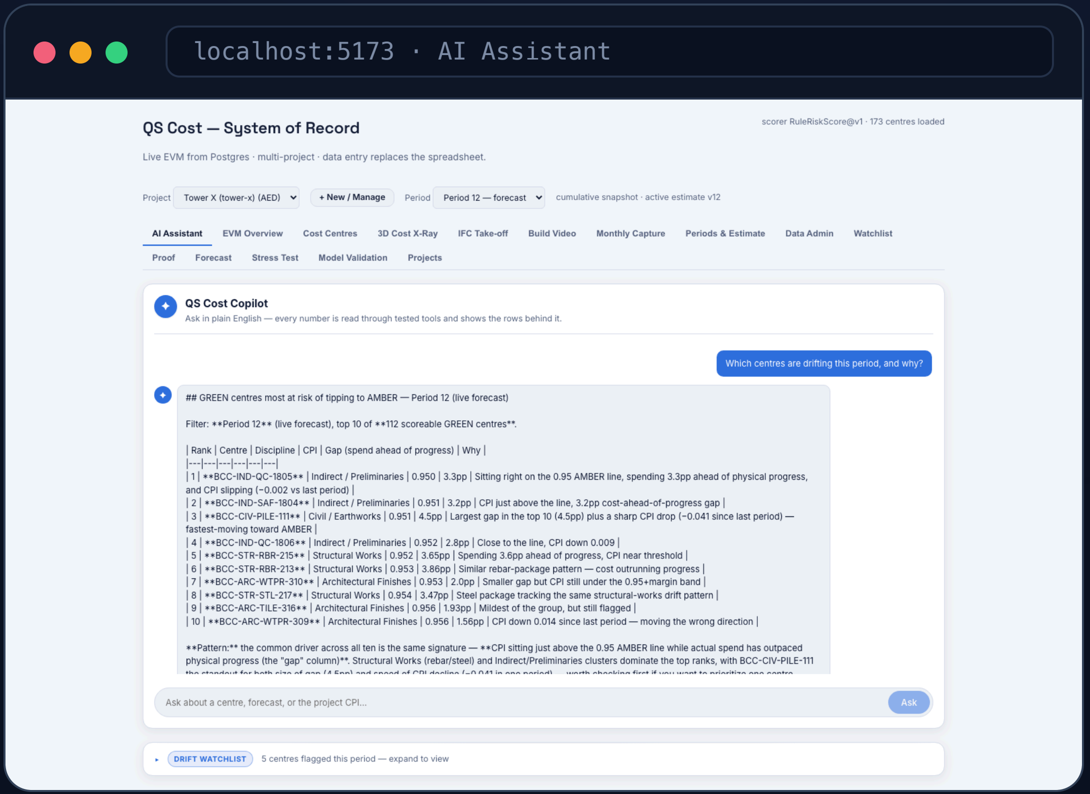


05 · ASK IT IN PLAIN ENGLISH

Tools compute. The model narrates.

- **13 read-only tools** running the same C# analytics the dashboard uses — the model has no database access and does no arithmetic.
- Every answer carries a **sources block** naming the tools called and the rows read.
- Tenancy is resolved **before** the model runs, so no prompt can reach another project.

Validated-versus-directional is enforced in the tools, not the prompt: the final-cost tool returns its "not validated" flag as part of the payload.

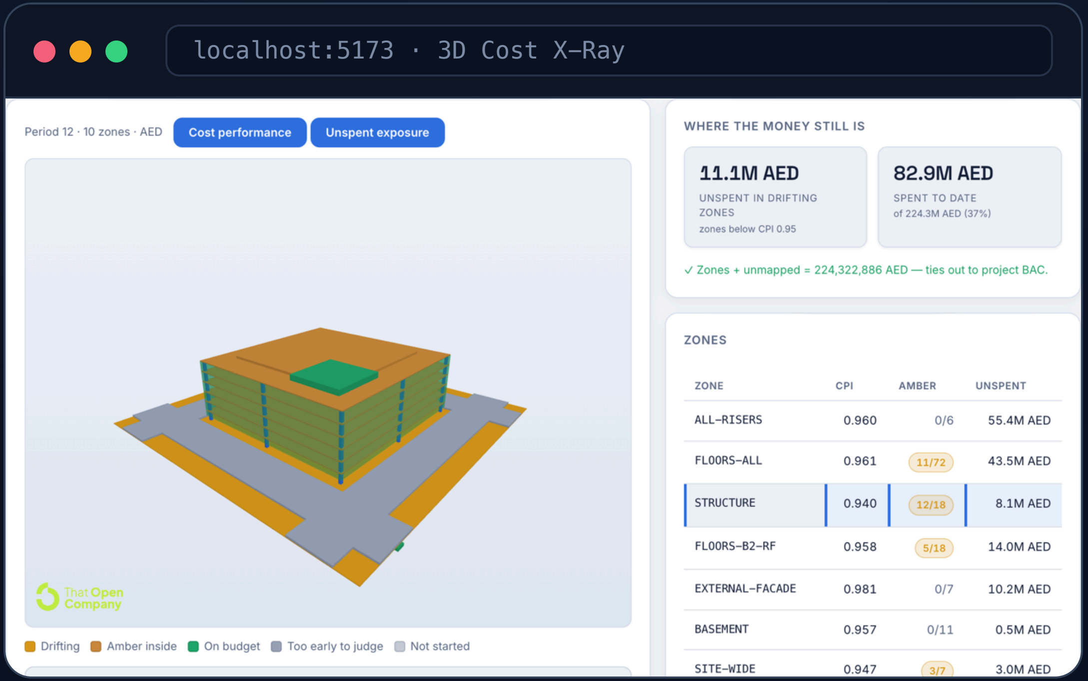


06 · ON THE BUILDING

The watchlist, on the building.

- Massing **derived from the bill** — every dimension carries the BOQ line it came from.
- **11.1M AED still unspent** in zones already below CPI 0.95.
- Zones + unmapped **ties out to project BAC**, exactly. Unlocated centres are shown, not dropped.

STRUCTURE has the worst ratio at CPI 0.940. FLOORS-ALL looks healthier at 0.961 — and has 43.5M AED still to lose.

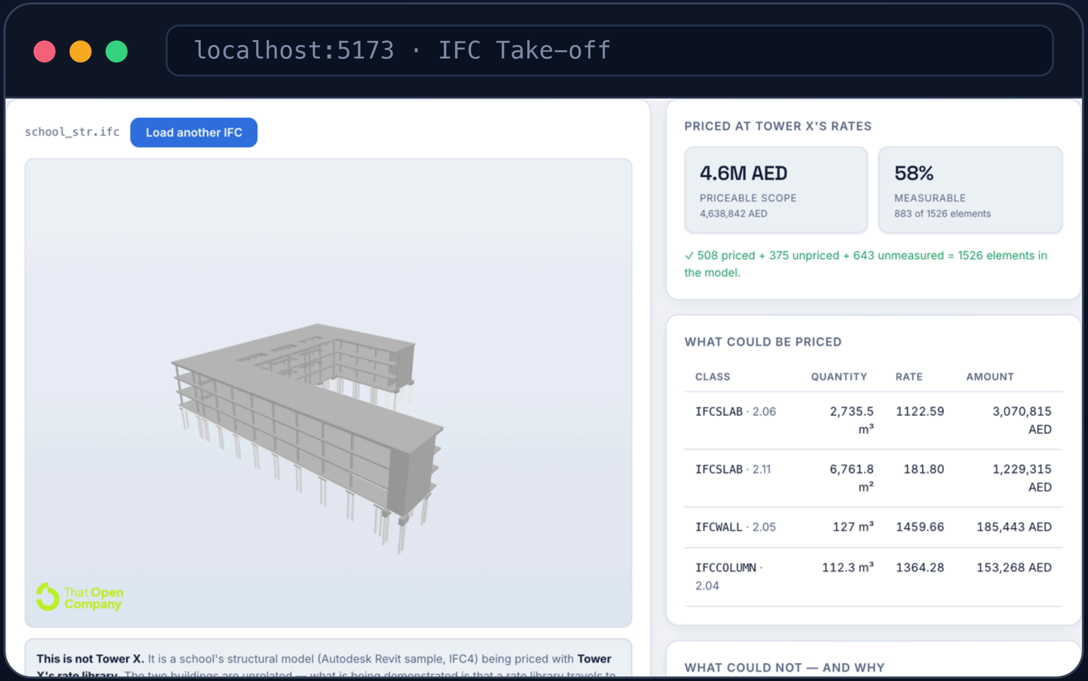


06 · ON THE BUILDING

Point it at any model. Get a price.

- **IFC Take-off** — a real Revit export, measured and priced with Tower X's rate library: **4.64M AED**, and **508 priced + 375 unpriced + 643 unmeasured = 1,526** elements.
- **Element register** — click an element, read the bill off it. 1,127 of 1,526 mapped, reaching 8 cost centres.
- **4D build sequence** — the sheet's progress curves, played on the model, coloured by alert. Rendered deterministically, byte-identical every run.

The 375 unpriced beams are the point: the bill prices no beam concrete, so they stay grey ghosts. The scope gap is visible as structure that never fills in.



07 · RUN IT AS A SYSTEM

Production-shaped, not a notebook.

Reporting workflow 16

Open and close periods, capture progress and cost as idempotent ledger deltas, publish estimate versions, rebaseline, cut over. Enforced by database procedures, not just the API.

Project management 17

Create or import a project from a workbook in one upload, with a reconciliation summary you can check. Re-import, rename, delete.

Data administration 18

Governed CRUD over 13 entities, grouped the way the source workbook is. The forms are generated from a server registry, so the screen cannot drift from what the server accepts.

Tenant isolation 19

PostgreSQL row-level security — the database refuses another tenant's rows even if the application asks. Authorisation runs before the Copilot's tools are built.

CREDIBILITY

Honest where it counts.

Validated

- Incremental spend — back-tested against four baselines on identical rows
- Estimate reconciliation — ties to the dirham, asserted on every build
- Variance tie-out — $\Sigma\ CV_r$ + residual = CV, exactly
- Take-off element count and the byte-identical video render

Directional — and labelled on screen

- Final cost / EAC — no ground truth at 13% median progress
- Variance attribution — uses estimate shares, names the evidence needed
- The IFC model is a school, not Tower X — mechanism, not valuation
- 4D build order is assumed; authentication is a development shim

Fourteen such claims, each with its reason, are listed in one place — the honesty ledger at the back of the feature guide.

UNDER THE HOOD

Four layers, one arithmetic.

Store `PostgreSQL 15`

Raw SQL schema, forced row-level security, security-invoker views, stored procedures for the reporting workflow. EVM identities computed in the database.

Analytics `C# / .NET 8`

Risk scorer, spend forecaster, stress tester, variance attributor, take-off pricer. Ridge regression, conformal intervals and the Cholesky solve are hand-written — **no ML dependency, no Python at run time**.

API `ASP.NET Core`

39 endpoints, 13 controllers. Every read of project data is authorised before it runs.

Interface `React 18 · three.js`

14 tabs, hand-built SVG charts, a WebGL viewer over real IFC, and a Copilot on Claude Sonnet through the Microsoft Agent Framework.

READ THE DETAIL

Nineteen features, one question each.

The feature guide gives every one of them a worked example from Tower X, the endpoint behind it, and what it cannot prove.

QS-Cost-Feature-Guide.pdf — 68 pages

tower-4d-build.mp4 — the build, played

docs/ — engineering deep dives